E Commerce Sales Performance Analysis

1. Introduction

This project presents an interactive dashboard developed using Power BI to analyze e-commerce sales performance. The dashboard converts raw sales data into meaningful visual insights that support business decision-making. It focuses on identifying revenue trends, store performance, and product-wise contribution across multiple regions.

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2. Objectives

The primary objectives of this project are:

- To analyze overall sales performance using key performance indicators (KPIs).
- To study monthly revenue trends from January to June 2025.
- To evaluate store-wise and product-wise sales contributions.
- To identify top revenue-generating products and regions.
- To examine relationships between sales metrics using correlation analysis.

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3. Tools and Technologies

The following tools and technologies were used in this project:

- Microsoft Corporation Power BI Desktop
- Power Query for data cleaning and transformation
- DAX (Data Analysis Expressions) for calculated measures
- Data visualization techniques

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4. Dataset Description

The dataset consists of structured e-commerce sales records collected from multiple stores. The main attributes include:

- Store Name
- Product Name
- Sales Revenue
- Month
- Order Volume
- Category

The data covers the period from **January to June 2025** and includes five stores: Bangalore, Chennai, Delhi, Kochi, and Mumbai.

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5. Data Preparation

Data preprocessing was carried out using Power Query in Power BI. The following steps were performed:

- Removal of missing and duplicate values
- Standardization of column names
- Conversion of data types for numerical and date fields
- Creation of calculated measures such as Total Revenue, Average Order Value (AOV), and Sales Volatility

These steps ensured accuracy and consistency of the dataset before visualization.

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6. Key Performance Indicators (KPIs)

The dashboard highlights the following KPIs:

- **Total Revenue:** $13.3 million
- **Order Volume:** 254
- **Average Order Value (AOV):** $52.36K
- **Maximum Sale Value:** $275K
- **Median Sale:** $22K
- **Sales Volatility:** $68.20K

These KPIs provide a summarized view of overall business performance.

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7. Dashboard Components and Visualizations

7.1 Revenue Trend Analysis

A line chart is used to display monthly revenue from January to June 2025. This visualization helps identify growth patterns, seasonal changes, and peak revenue months.

7.2 Revenue Distribution by Store and Product

Bar charts represent revenue distribution across various product categories such as Tablet, Mouse, Keyboard, Headphones, and Smart Watch. This allows comparison of product performance across stores.

7.3 Store-wise Revenue Contribution

A pie chart illustrates the percentage contribution of each store to total revenue. This visualization highlights the most and least contributing stores.

7.4 Top Revenue Generating Products

A ranked bar chart displays the products generating the highest revenue, helping identify best-selling items.

7.5 Store Performance and Product Breakdown

A matrix table presents detailed revenue values for each product across all stores. This enables comparative analysis between stores and product categories.

7.6 Regional Revenue Footprint

A map visualization shows the geographical distribution of revenue across Bangalore, Chennai, Delhi, Kochi, and Mumbai, providing spatial insights into regional performance.

7.7 Correlation and Advanced Analysis

Advanced analytical visuals include:

- Revenue density by category
- Variable correlation matrix
- Store-category connection map
- Monthly price-volume evolution
- 3D sales performance cube

These visuals help explore relationships among multiple sales metrics and uncover hidden patterns.

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8. Results and Insights

The analysis reveals the following key findings:

- Tablet and Mouse are the highest revenue-generating products.
- Bangalore and Delhi stores contribute significantly to total revenue.
- Revenue shows a steady upward trend from January to June.
- Order volume has a strong positive correlation with total revenue.
- Certain categories demonstrate higher revenue stability than others.

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9. Discussion

The interactive nature of the dashboard allows users to filter and explore data dynamically. Store-wise, product-wise, and monthly analyses can be performed efficiently. The inclusion of correlation and advanced analytical visuals improves the depth of business insights and supports strategic planning.

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10. Conclusion

This project demonstrates the effective use of Power BI for comprehensive e-commerce sales analysis. The dashboard transforms complex datasets into visually meaningful information and supports informed decision-making by identifying trends, top performers, and relationships between key metrics.

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11. Future Scope

Future enhancements may include:

- Integration of profit and cost analysis
- Addition of slicers for year, store, and category
- Sales forecasting using predictive analytics
- Customer segmentation and behaviour analysis